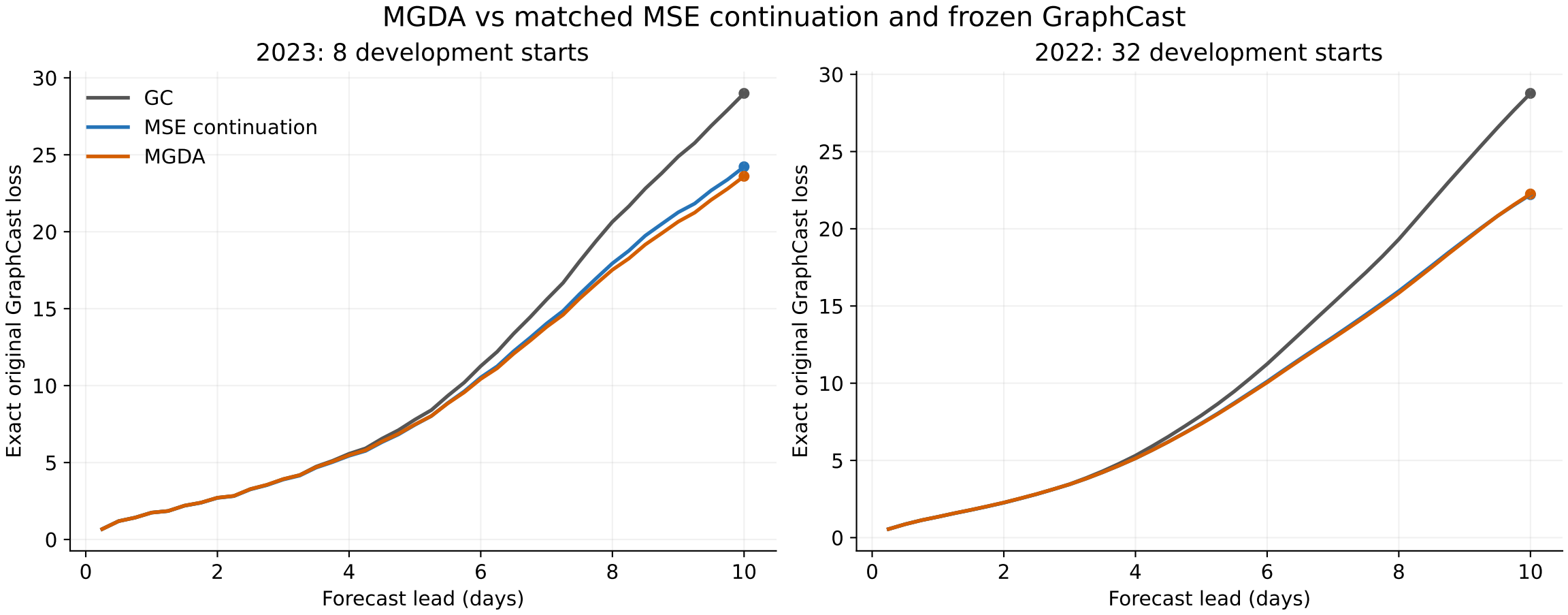
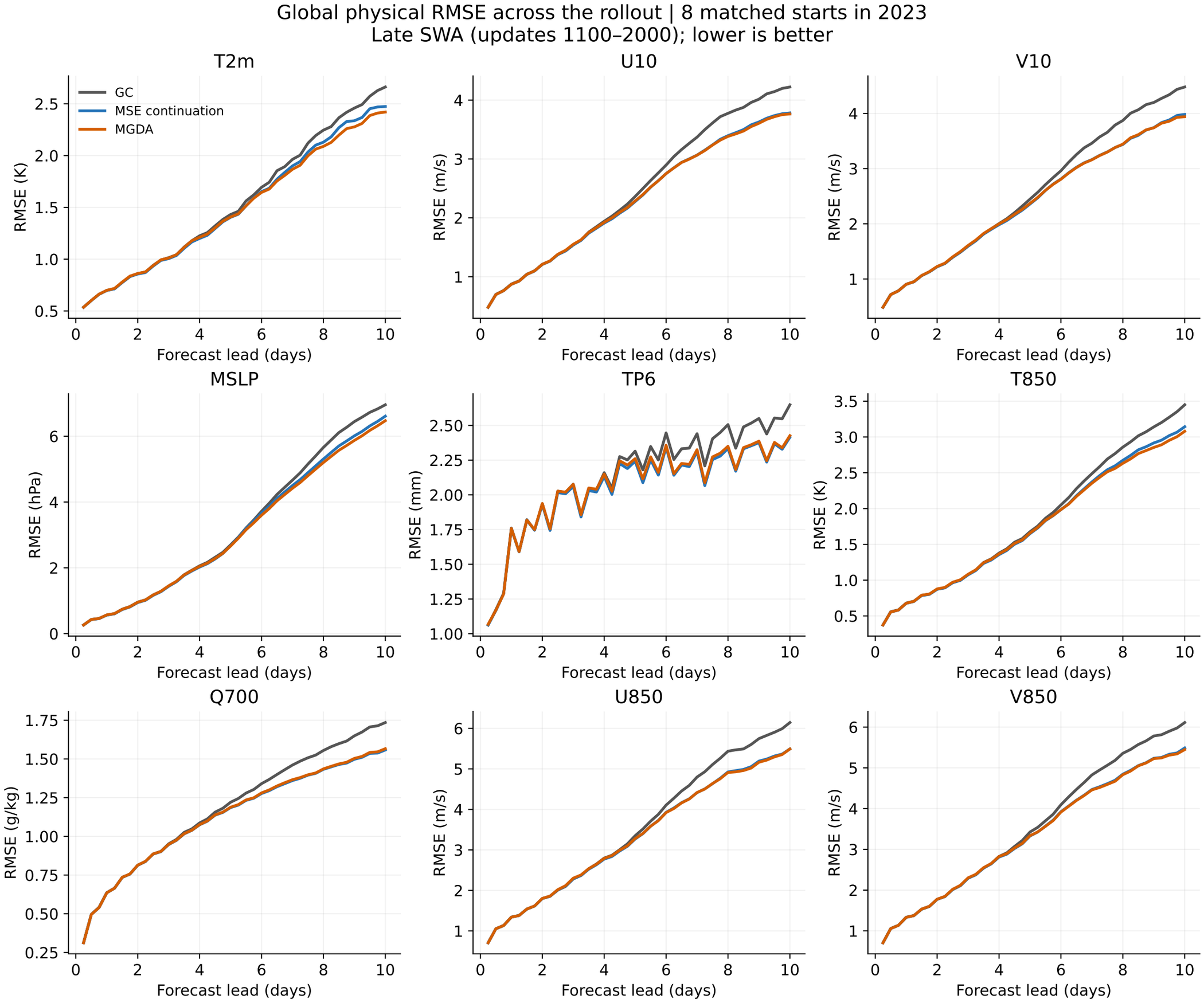
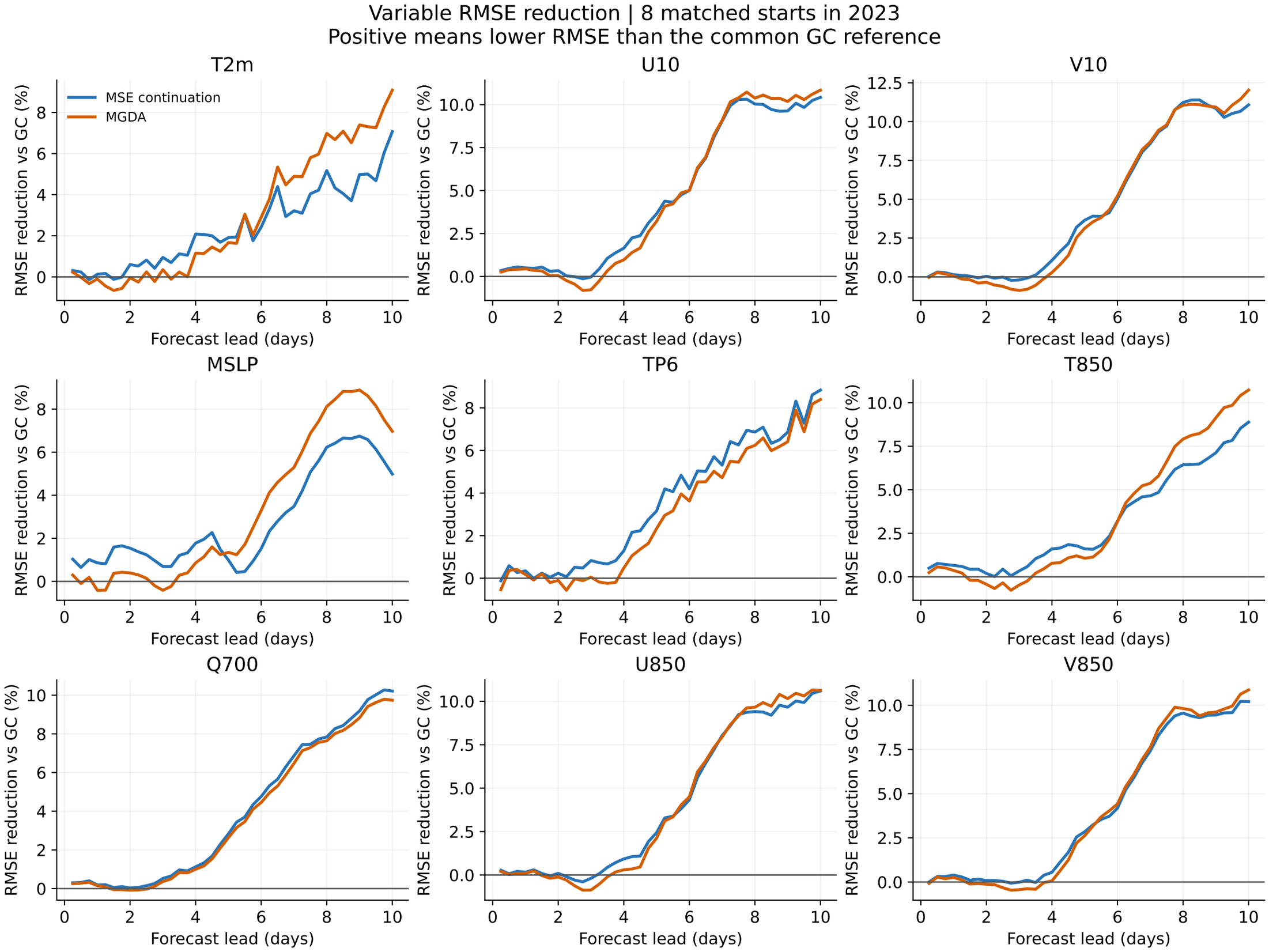

MGDA vs matched MSE continuation and frozen GraphCast
2023: 8 development starts
2022: 32 development starts
GC
MSE continuation
MGDA
Exact original GraphCast loss
Forecast lead (days)
0
5
10
15
20
25
30
0
2
4
6
8
10


Global physical RMSE across the rollout | 8 matched starts in 2023
Late SWA (updates 1100–2000); lower is better
T2m
U10
V10
MSLP
TP6
T850
Q700
U850
V850
GC
MSE continuation
MGDA
RMSE (K)
RMSE (m/s)
RMSE (hPa)
RMSE (mm)
RMSE (g/kg)
Forecast lead (days)


Variable RMSE reduction | 8 matched starts in 2023
Positive means lower RMSE than the common GC reference
T2m
U10
V10
MSLP
TP6
T850
Q700
U850
V850
MSE continuation
MGDA
RMSE reduction vs GC (%)
Forecast lead (days)